ASSIGNMENT 1 FRONT SHEET

Qualification	**BTEC Level 5 HND Diploma in Computing**		
Unit number and title	Unit 06: Managing a Successful Computing Project		
Submission date	20/10/2022	**Date Received 1st submission**	
Re-submission Date		**Date Received 2nd submission**	
Student Name	Le Hoang Du	**Student ID**	GCC200416
Class	GCC0901	**Assessor name**	Nguyen Trung Viet
Student declaration I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.			
		Student's signature	Le Hoang Du

Grading grid

P1	P2	P3	P4	M1	M2	D1

✡ **Summative Feedback:**		✡ **Resubmission Feedback:**
Grade:	**Assessor Signature:**	**Date:**
IV Signature:		

ASSIGNMENT 1 BRIEF

Qualification	**BTEC Level 5 HND Diploma in Computing**		
Unit number and title	Unit 06: Managing a Successful Project		
Assignment title	Plan and conduct a small scale research activity		
Academic Year	2022 - 2023		
Unit Tutor	Nguyen Trung Viet		
Issue date		**Submission date**	20/10/2022

Submission Format:

Format: The submission is in the form of an individual written report that shows how you have manage the project. This should be written in a concise, formal business style using single spacing and font size 12. You are required to make use of headings, paragraphs and subsections as appropriate, and all work must be supported with research and referenced using the Harvard referencing system. Please also provide a bibliography using the Harvard referencing system.

Submission Students are compulsory to submit the assignment in due date and in a way requested by the Tutors. The form of submission will be a soft copy in PDF posted on corresponding course of http://cms.greenwich.edu.vn/

Note: The Assignment *must* be your own work, and not copied by or from another student or from books etc. If you use ideas, quotes or data (such as diagrams) from books, journals or other sources, you must reference your sources, using the Harvard style. Make sure that you know how to reference properly, and that understand the guidelines on plagiarism. *If you do not, you definitely get fail*

Assignment Brief and Guidance:

Introduction to theme

The environmental impact of digital transformation

The amount of data created and stored globally is expected to reach 175 Zettabytes by 2025, a six-fold increase from 2018. This will demand additional hardware and power consumption, which; in turn, will increase the environmental impact of the digital sector and there is already increasing attention on the environmental footprint of ICT equipment and services as they become more widespread in all aspects of human life.

It is the responsibility of everyone to take action in addressing the challenges of climate change, as professionals we must also seek ways that the digital sector can play its part. While digital technologies are one of the sectors that has achieved greater efficiency; achieving about 100 times more computation power from the same amount of energy per decade, it remains unsustainable. The sector must continue to seek ways in which it can continue to support and drive innovation, while addressing the global climate emergency for a greener and fairer future.

This unit will enable students to explore the impact of digital endpoint devices and ways to reduce environmental damages, OR the potential of refurbishing, repairing and reusing digital devices rather than replacing

Tasks

As a member of Research and Development department, you have been assigned a mini-project to find out the impact of digital endpoint devices and ways to reduce environmental damages, OR the potential of refurbishing, repairing and reusing digital devices rather than replacing.

You need to do primary research (both qualitative and quantitative research) and secondary research to find out that impact and conduct a report for your research. Even it's a mini-project, you must apply project management (PM) techniques such as project charter with aims, objectives, cost etc. As for time management, you need to produce WBS and Gantt chart with reasonable tasks and time. A project logbook is required to provide evidence of the project development process and ongoing reflection for every week. This logbook will be needed later for your reflection and evaluation in Assignment 2. As part of QA (quality assurance) policy, in the report you also need to critically evaluate the PM process and appropriate research methodologies applied.

Your report must have an introduction stating the project aims and objectives. This must be followed by a copy of your project management plan. Your plan should show the milestones when you will review with your tutor your ongoing progress so far. You will submit your logbook which shows how you have carried out the project.

<table>
<tr><th colspan="3">Learning Outcomes and Assessment Criteria</th></tr>
<tr><th>Pass</th><th>Merit</th><th>Distinction</th></tr>
<tr><td colspan="2">LO1 Establish project aims, objectives and timeframes based on the chosen theme</td><td rowspan="2">LO1 & LO2
D1. Critically evaluate the project management process and appropriate research methodologies applied.</td></tr>
<tr><td>P1 Devise project aims and objectives for a chosen scenario.
P2 Produce a project management plan that covers</td><td>M1 Produce a comprehensive project management plan, milestone schedule and project schedule for monitoring and completing the aims and objectives of the project.</td></tr>
</table>

aspects of cost, scope, time, quality, communication, risk and resources. **P3** Produce a work breakdown structure and a Gantt Chart to provide timeframes and stages for completion.		
LO2 Conduct small-scale research, information gathering and data collection to generate knowledge to support the project		
P4 Carry out small-scale research by applying qualitative and quantitative research methods appropriate for meeting project aims and objectives	**M2** Evaluate the accuracy and reliability of different research methods applied.	

Table of Contents

1 Project Management 8
1.1 Introduction 8
1.2 Project initialization 10
1.3 Project Management Plan 11
1.4 Planning 20
1.5 Produce a comprehensive project management plan, milestone schedule and project schedule for monitoring and completing the aims and objectives of the project 23
2 Research 27
2.1 Primary Reasearch 27
2.2 Secondary research 30
3 References 34

List of figures

Figure 1: WBS 21
Figure 2: CPM 27

List of tables

Table 1: Risk management 13
Table 2: Cost estimation 16
Table 3: Task and cost table 18
Table 4: Monitoring progress 25
Table 5: Qualitative vs Quantitative 28
Table 6: Survey questions 29
Table 7: List of articles 31

1 Project Management

1.1 Introduction

1.1.1 Title of the project

The impact of digital storage devices on the environment: analyzing and proposing solutions.

1.1.2 Introduction

Along with the development of human life, the methods of storing information and data are also changed and developed over time. In the early days, people used paper to store data, then the digital transformation trend arose, resulting in storage devices such as hard drives, floppy disks, CDs/DVDs, and USB drives being developed and birth bringing increasing storage capacity and faster data processing speed. However, the development of human technical life is more and more powerful, leading to an increasingly large amount of data generated. The amount of data created and stored globally is expected to reach 175 Zettabytes by 2025, a six-fold increase from 2018. This will demand additional hardware and power consumption.

Therefore, popular storage devices such as hard drives or USB drives are not meeting human needs. Accordingly, cloud storage services were born and growing strongly, they can provide a huge and possibly unlimited storage capacity, along with extreme processing speed. Fast and accessible anywhere. The strong development of cloud storage and digital transformation services makes the use of storage devices less and less, leading to them being neglected and potentially turning into electronic waste that affects the environment. To reduce the harmful effects of digital storage devices on the environment, we can do the following:

- Reuse digital storage devices can use them for other purposes such as backup data storage, etc.
- Collect and use parts of such storage devices such as memory cards and microchips to produce other more useful devices instead of abandoning them.
- Donate used digital storage devices to social programs, children's safety initiatives, environmental causes, and more.

This report explores the impact of specific digital endpoint storage devices (such as hard drives, USB drives, etc), and offers ways to reduce their environmental damage. The report will include primary research (both qualitative and quantitative research) and secondary research to find out the impact of these devices on the environment. In addition, the report also mentions aims, objectives, costs, important milestones in the project implementation process, etc. Finally, the report also provides the project's WBS and Gantt chart.

1.1.3 Project Management

Project management is the discipline of successfully guiding a project from inception to completion utilizing predetermined concepts, processes, and regulations. Project management often shortened as PM involves using these guidelines and techniques as well as tools and technology to make sure that a project can be finished in a way that fulfills all specified outcomes, from budgetary constraints to end-goal goals. The desired end is to produce the result on time and within budget, and the project management plan is supposed to successfully and efficiently lead all areas of a project from start to completion (Mary K. Pratt, 2018).

1.1.4 Software Project

A software project is the whole process of developing software, from gathering requirements through testing and maintenance, carried out in accordance with execution techniques over a predetermined amount of time to produce the desired software output. A software development project is a complex undertaking by two or

more persons within the boundaries of time, budget, and staff resources that produce new or enhanced computer code that adds significant business value to a new or existing business process (Tuorialspoint, 2022).

1.1.5 Project Management in software engineer

An appropriate method for organizing and managing software projects is called software project management (SPM). Software project planning, implementation, monitoring, and control are all done as part of project management. Software developers use project management skills to develop the requirements for the software, identify and track the product development tasks, communicate within the development team and with clients, test cases, and manage quality, the schedule, and resources (staff, equipment, labs, and more). A business must supply high-quality goods, manage costs within the client's budgetary limits, and complete projects on time. Thus, it is essential for software project management to take into account user needs as well as financial and scheduling restrictions (Adrienne Watt, 2014).

1.1.6 Project inception

Project Inception is the initial stage of the majority of projects where the purpose of the project is explored, a solution for implementation is identified, benefits that will result from a successful completion are defined, the time needed to complete the project is estimated, and funding requests are made. The goal is to start defining the project's broad characteristics and creating the proper management environment needed to start and maintain the project. A project's scope of work, general guidelines, and overview are all defined in a project's initiation document, which may be used to discuss the project's requirements with the team. It serves as an important document for the internal team and external stakeholders and makes it easier to communicate the intent and set the right course for the project (Imagine Projects, 2021).

1.1.7 Project Planning

In project management, the creation of the necessary paperwork is a procedural stage called "project planning" that ensures the effective completion of the project. All procedures necessary to create, prepare, integrate, and coordinate new plans are documented. How the project is carried out, tracked, managed, and closed is made crystal apparent in the project plan (Techopedia, 2012).

Project planning requires an in-depth analysis and structuring of the following activities:

- Setting project goals
- Identifying project deliverables
- Creating project schedules
- Creating supporting plans

1.1.8 Project execution

The phase of the project life cycle known as project execution is when the work is completed and every aspect of the project plan is put into effect. Project planning and scheduling are only the start. The majority of a project's time and labor is spent on its implementation. As a result, project management execution is very crucial. The work is completed during the project's execution phase. Workflow remains efficient when task owners start their work and the project manager monitors that it is finished on time. As the project moves forward, problems will inevitably develop that call for swift modifications, therefore project monitoring and control are crucial components of this phase (Adeaca Corp, 2021).

1.1.9 Closing a project

Project closing is the last phase of a project. This phrase is used by project managers to denote that a client, stakeholder, or customer has authorized the usage of project deliverables. The project or product has to have a maintenance strategy in place. Any endeavor's management will undertake a comprehensive evaluation of the project to guarantee a successful ending. The management team will contribute to this phase by making observations and comments that will be summarized in a "lessons learned" paper. This acts as a guide for upcoming projects. A project isn't only closed when the documentation is finished, any vendor contracts are finished, and the team is free to concentrate on other projects. A project's end-of-project review ensures that all of the project's goals have been completed and that any remaining concerns, such as risk, have been resolved (Simplilearn, 2022).

1.2 Project initialization

According to Adrienne Watt (2014), the project initialization phase, which involves beginning a new project, is the initial stage of the project management life cycle. The identification of the business opportunity or problem, the definition of the solution, the formation of the project, and the appointment of the project team to develop and deliver the solution to the client are all done during the beginning phase. A detailed definition of the issue or opportunity is provided in a business case, along with a recommendation for the best course of action. The business case includes:

- A detailed description of the problem or opportunity with headings such as Introduction, Business Objectives, Problem/Opportunity Statement, Assumptions, and Constraints
- A list of the alternative solutions available
- An analysis of the business benefits, costs, risks, and issues
- A description of the preferred solution
- Main project requirements
- A summarized plan for implementation that includes a schedule and financial analysis

1.2.1 Main aim of the project

The project will aim to explore the impact of specific digital storage devices (hard drives, USB drives, etc) on the environment during digital transformation at the ABC company and offer solutions to reduce their environmental damage.

1.2.2 List of Objectives to achieve the aim

- Understand exactly what digital endpoint storage devices are, and collect information about those devices' effects on the environment through articles or well-known institutions such as WWF, WHO, or universities.
- Open surveys to poll people about the impact of digital storage devices on the environment to collect data to support the project.
- Explore and find out the impact of specific digital endpoint storage devices such as hard drives, USB drives, etc on the environment during digital transformation at the ABC company.
- Offer effective solutions to reduce the damage caused by digital storage devices to the environment.
- This project should be completed by December 1, 2022 which is the deadline for the project.

- Ensuring most employees of ABC company recognized the project's effectiveness for environmental protection in the impact of digital transformation after the project was completed and announced.
- Provide user documentation and train employees of the ABC company how to use cloud service storage.

1.3 Project Management Plan

A bundle of documents known as a project management plan describes the how, when, and what-ifs of a project's execution. It provides an overview of the value proposition of the project, the stages necessary in its execution, available resources, communication tools and protocols, risks, stakeholders (and their responsibilities), and the deliverables needed to complete the project. An executive summary, team and Gantt charts, a risk assessment, and sub-plans for resource and communication management are all included in the documents (Cassie Bottorff & Alana Rudder, 2022).

1.3.1 Scope

In project management, the term "scope" describes the kind and range of project operations. Setting boundaries for what is considered important and irrelevant to the project is the process of defining scope. Project managers can plan a project team, objective, budget, and other elements in addition to the scope (Indeed Editorial Team, 2020).

The following is the scopes of our project:

Project scope description: The management of ABC company assigned the Research and Development department to conduct a mini-project to find out the impact of digital storage devices on the environment during digital transformation at the ABC company and propose solutions to reduce their damage to the environment. The Research and Development department must provide a complete report of the research including primary and secondary research must be provided along with the project management techniques to be applied.

Project acctepance criteria:

- Complete the project on time.
- Provides accurate research and reports on the impact of digital storage devices during digital transformation on the environment at ABC company.
- The proposed solutions to reduce the damage of digital storage devices to the environment are 90% or more effective.
- The management board of ABC company approves and recognizes the effectiveness of the project after completion.
- Most employees of ABC company recognized the effectiveness of the project.
- The cost from project implementation to completion does not exceed the initial estimated budget of the project.
- The implementation cost of the project does not exceed 12.500 USD.
- Training employees of the ABC company how to use cloud service storage.

Project exclusions:

- The data and research were conducted in the ABC company's region, so it is not certain that it applies to other regions.

- No guarantee that the effectiveness of solutions to reduce the damage of digital storage devices will be 100% after implementation.
- Not responsible for data errors after the time the project has been completed.

Project constraints:

- Impact of the Covid-19 pandemic.
- Affected by natural disasters and weather such as storms and floods.
- The project implementation budget is not enough.
- The project should be completed by December 1, 2022.
- The project team members will be backfilled as necessary.
- Time/Resource/personel limitaions.

Project assumptions:

- All team members have digital transformation knowledge and all required skills.
- Permanent members will not be reassigned.
- Funding will be available when required.
- All equipment is in good condition.
- The supplier will deliver consumables on time.

1.3.2 Time

Project time management is the process of managing a project that involves reviewing the work and creating a timeline to make sure you finish it on time. The management of time spent and project task and activity progress is known as time management. Planning, scheduling, monitoring, and managing all project operations are necessary for effective time management. According to the Project Management Institute, one of the six key functions of project management is time management. Some individuals also refer to the methods and tools for managing time when they talk about project time management (Shawn Dickerson, 2021).

This project will be started from September 15, 2022, to December 1, 2022. Details of the project implementation schedule will be presented in table 3 below.

1.3.3 Communication

The goal of communications management is to keep everyone informed. Determining the sorts of information you will convey, who will get it, how it will be communicated, and when it will be released and distributed are all part of the communications planning process. It turns out that 90% of a project manager's time is spent communicating, so it's critical to ensure that everyone receives the proper information at the appropriate time. The communication plan documents the types of information the stakeholders need, when the information should be distributed, and how the information will be delivered. The types of information you will communicate typically include project status, project scope statements and updates, project baseline information, risks, action items, performance measures, project acceptance, and so on (Adrienne Watt, 2014).

Our project will require at least 3 key meetings during project planning:

- The first meeting will take place between the management of ABC company and the project team to set objectives and directions for the project, and collect the requirements that the management wants. In addition, the meeting will discuss necessary factors for the project such as budgets, deadlines, etc.
- The second meeting will take place between members of the project team including the project manager. This meeting includes activities such as proposing methods to achieve goals, dividing tasks, setting deadlines for each member, discussing the topic of digital transformation, etc.
- The third meeting will be held again among team members to summarize and re-evaluate the entire project after completion. From there, the lessons learned and the unsatisfactory aspects of the project can be drawn to improve performance for the next project.

In addition to the above 3 main meetings, if any problems arise during project implementation, individuals in the team will meet face-to-face to solve the problem as quickly as possible so as not to cause any problems affecting the progress of the entire project.

1.3.4 Risks

Any unpredictable circumstance or incident that might have an impact on your project is a risk. Risks are not always bad. Your project may benefit from particular occurrences (like discovering a simpler technique to do a task) or circumstances (like decreased pricing for specific resources). When this occurs, we refer to it as an opportunity, but it is still treated the same way as a danger. A project can encounter risks at any point. Some of them are connected to specific activities, while others come from outside the project and might appear suddenly. In general, a risk event that happens at the end of a project might be more expensive in terms of time and money than a similar event that occurs closer to the beginning of the project. That is because as time passes there will be a greater value of work in progress and higher sunk costs at the risk of loss or damage (Dennis Lock, 2017).

According to Christian W. Dawson (2009), inextricably connected to project management, risk management operates concurrently with and adheres to a very similar workflow to project management. In the same way that project management is creating a strategy for the project and managing it using that plan as it advances, risk management entails identifying risks at the commencement of the project and managing those risks as it proceeds. You will be implementing the steps of the risk management process as you go through the project management phases. To help you manage and control risks in your own project, we'll provide a risk management approach in this part. The four main stages of this risk management process are:

- Identify risks
- Assess the impact of risks
- Alleviate critical risks
- Control risks

Following is the risk management table of our project, which will list the possible risks, their classification and suggested solutions for them:

Table 1: Risk management

Name of risk	Category Risk	Priority	Impact	Suggest action
Project aims and objectives is not well-defined	Technical Risk	**High**	The project was implemented in the wrong direction	Conduct meetings to discuss project goals and revise

			and did not meet the requirements of the management board of ABC company.	relevant documents such as Project Charter, PID.
Project schedule is not clearly defined or understood	Schedule Risk	**Medium**	Tasks will not be clearly divided among members, deadlines and milestones in the project are not clearly defined affecting the progress of the project.	Hold scheduling workshops with the project team so they understand the plan and the likelihood of missed tasks is reduced.
Poor experience of the project manager in the relation with the team members and with the management of the ABC company	Human Resource Risk	**Medium**	Affect the communication process between members, thereby affecting project activities.	It is necessary that the project manager should be experienced personal in project management or should have experience in monitoring and coordinating activities.
Selecting inappropriate research methods	Technical Risk	**High**	Making the project implementation ineffective and not meeting the set objectives.	Hold a discussion among team members to jointly analyze research methods to find the most suitable research method for the project.
Team members lack of expertise knowledge about digital transformation	Technical Risk	**High**	Failing to meet the assigned tasks, affecting the quality of the project.	The project manager must select members with full knowledge of digital transformation or open knowledge training courses for members.

The assigned tasks are not completed on time	Schedule Risk	**Medium**	Affects the progress of the project and may cause the project to not be completed on time	The project manager must create the initial schedule carefully and clearly, and proceed to ask members to overtime when the work is not on the schedule.
The conflict between team members during project implementation	Human Resource Risk	**High**	Affecting the process of communication and information exchange for project implementation among team members, making project implementation ineffective.	The project manager acts as a mediator when there is a conflict in the group, holding discussion meetings to help team members understand each other's points of view.
Lack of necessary funding or resources	Budget Risk	**Low**	Making the project implementation inefficient	Do thorough research and analysis of project essentials before starting the project, including contingencies should problems arise
Impact due to obstacles caused by the Covid-19 pandemic or natural disasters	External Risk	**Low**	Team members face obstacles because they cannot meet face-to-face to exchange information and work together	Implement forms of remote work through online meeting platforms like Google Meet or Zoom

1.3.5 Resources

Resources include any necessary people, equipment, locations, or sums of money to carry out all the planned operations. There must be resources allotted to each action on your to-do list. You must be aware of the resources' availability before you may allocate them to your project. When it comes to resource availability,

you should know what resources are accessible, when you may utilize them for your project, and under what circumstances. Remember that some resources, such as consultants or training facilities, need advanced scheduling and may only be accessible during specific hours. Before you can complete your project's planning, you must be aware of this. You'll also need the activity list that you created earlier, and you'll need to know how your organization typically handles resources. Once you've got a handle on these things, you're set for resource estimation (Adrienne Watt, 2014).

The following are the necessary resources for our project:

- Material
  - Printers
  - Computers and laptops
  - Smartphones
  - Digital storage devices (USB drives, hard drives, etc)
  - Cloud Service Premium accounts
  - Other necessary devices
- Human
  - Le Hoang Du: project manager
  - Nguyen Ngoc Hai: developer
  - Nguyen Minh Nghia: developer
  - Truong Phat Dat: developer
- Cost: 12.500 USD

1.3.6 Cost estimation

Cost estimation in the context of project management is the process of calculating the expenses related to finishing a project on schedule and within the project's scope. In the early phases of project planning, initial, high-level estimates are frequently employed and can affect whether a project is eventually pursued or not. More precise and granular cost estimates are required once a project is authorized and an organization decides to move forward with it in order to properly deploy various resources. The direct and indirect expenses related to carrying out a project should both be included in an accurate cost estimate. Depending on the specifics of the venture, this will likely include various overhead costs (utilities, labor, etc.), labor costs (including both time and wages paid), materials and equipment costs, vendor fees (if the project requires third-party workers, freelancers, or other contractors), and more (Tim Stobierski, 2019).

The following table shows cost estimates of project resources including human, material, and cost:

Table 2: Cost estimation

No.	Resource Name	Type	Std. Rate	Ovt. Rate	Cost/Use
1	Du	Work	$15.00/hr	$20.00/hr	$0.00
2	Hai	Work	$10.00/hr	$15.00/hr	$0.00
3	Dat	Work	$10.00/hr	$15.00/hr	$0.00

4	Nghia	Work	$10.00/hr	$15.00/hr	$0.00
5	Printer	Material	$0.00	$0.00	$100.00
6	Computer	Material	$0.00	$0.00	$200.00
7	Laptop	Material	$0.00	$0.00	$150.00
8	Smartphone	Material	$0.00	$0.00	$100.00
9	Storage device	Material	$0.00	$0.00	$150.00
10	Cloud Service account	Material	$0.00	$0.00	$200.00
11	Other cost	Cost	$0.00	$0.00	$640.00

Table 3: Task and cost table

No.	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Cost
1	**Digital Storage Devices Project**	**56 days**	**Thu 9/15/22**	**Thu 12/1/22**			**$12,500.00**
2	**Initiation**	**11 days**	**Thu 9/15/22**	**Thu 9/29/22**			**$3,460.00**
3	Project Plan	1 day	Thu 9/15/22	Thu 9/15/22		Du	$120.00
4	Scope Statement	2 days	Fri 9/16/22	Mon 9/19/22	3	Dat, Du, Hai, Nghia	$720.00
5	Schedule plan	1 day	Tue 9/20/22	Tue 9/20/22	4	Nghia	$80.00
6	Communication plan	1 day	Wed 9/21/22	Wed 9/21/22	5	Dat, Nghia	$160.00
7	Risk plan	2 days	Thu 9/22/22	Fri 9/23/22	6	Du, Nghia	$400.00
8	Resources plan	1 day	Mon 9/26/22	Mon 9/26/22	7	Hai, Dat, Cloud Service account[1], Computer[1], Laptop[1], Printer[1], Smartphone[1], Storage device[1], Other cost[1]	$1,700.00
9	Budget plan	1 day	Tue 9/27/22	Tue 9/27/22	8	Nghia	$80.00
10	WBS	1 day	Wed 9/28/22	Wed 9/28/22	9	Dat	$80.00
11	Gantt Chart	1 day	Thu 9/29/22	Thu 9/29/22	10	Du	$120.00
12	**Research**	**13 days**	**Fri 9/30/22**	**Tue 10/18/22**			**$2,680.00**
13	Find document (articles/books)	2 days	Fri 9/30/22	Mon 10/3/22	11	Hai, Nghia	$320.00

14	Analyze document	2 days	Tue 10/4/22	Wed 10/5/22	13	Dat, Du, Hai	$560.00
15	Primary research	3 days	Thu 10/6/22	Mon 10/10/22	14	Du, Hai	$600.00
16	Secondary research	3 days	Tue 10/11/22	Thu 10/13/22	15	Dat, Nghia	$480.00
17	Write report	3 days	Fri 10/14/22	Tue 10/18/22	16	Dat, Hai, Nghia	$720.00
18	**Implementation**	**20 days**	**Wed 10/19/22**	**Tue 11/15/22**			**$4,520.00**
19	Buy digital storage devices	2 days	Wed 10/19/22	Thu 10/20/22	17	Dat, Hai	$320.00
20	Buy Cloud Service Premium accounts	1 day	Fri 10/21/22	Fri 10/21/22	19	Du	$120.00
21	Research and report on digital transformation	3 days	Mon 10/24/22	Wed 10/26/22	20	Du, Nghia	$600.00
22	Compare digital storage devices and cloud services	3 days	Thu 10/27/22	Mon 10/31/22	21	Dat, Hai, Nghia	$720.00
23	Explore the impact of digital storage devices	3 days	Tue 11/1/22	Thu 11/3/22	22	Du, Nghia	$600.00
24	Offer ways to reduce environmental damage	3 days	Fri 11/4/22	Tue 11/8/22	23	Dat, Du, Hai	$840.00
25	Deploy cloud storage services for employees	2 days	Wed 11/9/22	Thu 11/10/22	24	Dat, Du, Hai, Nghia	$720.00
26	Write report	3 days	Fri 11/11/22	Tue 11/15/22	25	Du, Nghia	$600.00
27	**Closing Project**	**12 days**	**Wed 11/16/22**	**Thu 12/1/22**			**$1,840.00**
28	Check and evaluate the whole project	2 days	Wed 11/16/22	Thu 11/17/22	26	Du, Hai	$400.00

29	Summarize into a complete report	4 days	Fri 11/18/22	Wed 11/23/22	28	Nghia	$320.00
30	Project report and presentation	2 days	Thu 11/24/22	Fri 11/25/22	29	Dat, Hai, Nghia, Du	$720.00
31	User docmentation	2 days	Mon 11/28/22	Tue 11/29/22	30	Du	$240.00
32	Closing documentation	2 days	Wed 11/30/22	Thu 12/1/22	31	Dat	$160.00

1.4 Planning

1.4.1 WBS

Work may be made more manageable and accessible by using a common productivity strategy called task breaking. The Work Breakdown Structure (WBS), one of the most crucial papers in project management, is the instrument that applies this method to projects. It incorporates baselines for scope, cost, and time on its own, ensuring that project plans are in sync. The usage of work breakdown structures (WBSs) allows you to precisely identify the work that needs to be done to accomplish your project by breaking it down into ever-lower levels of detail. You should start a WBS by decomposing your project into the key goals you determined during the project definition. You might only be able to break your project down into two or three main areas of work or you might be able to identify several broad areas of activity (Christian W. Dawson, 2009).

The following is the WBS table of our project, which consists of 4 main phases which are Project Management, Research, Execute, Close:

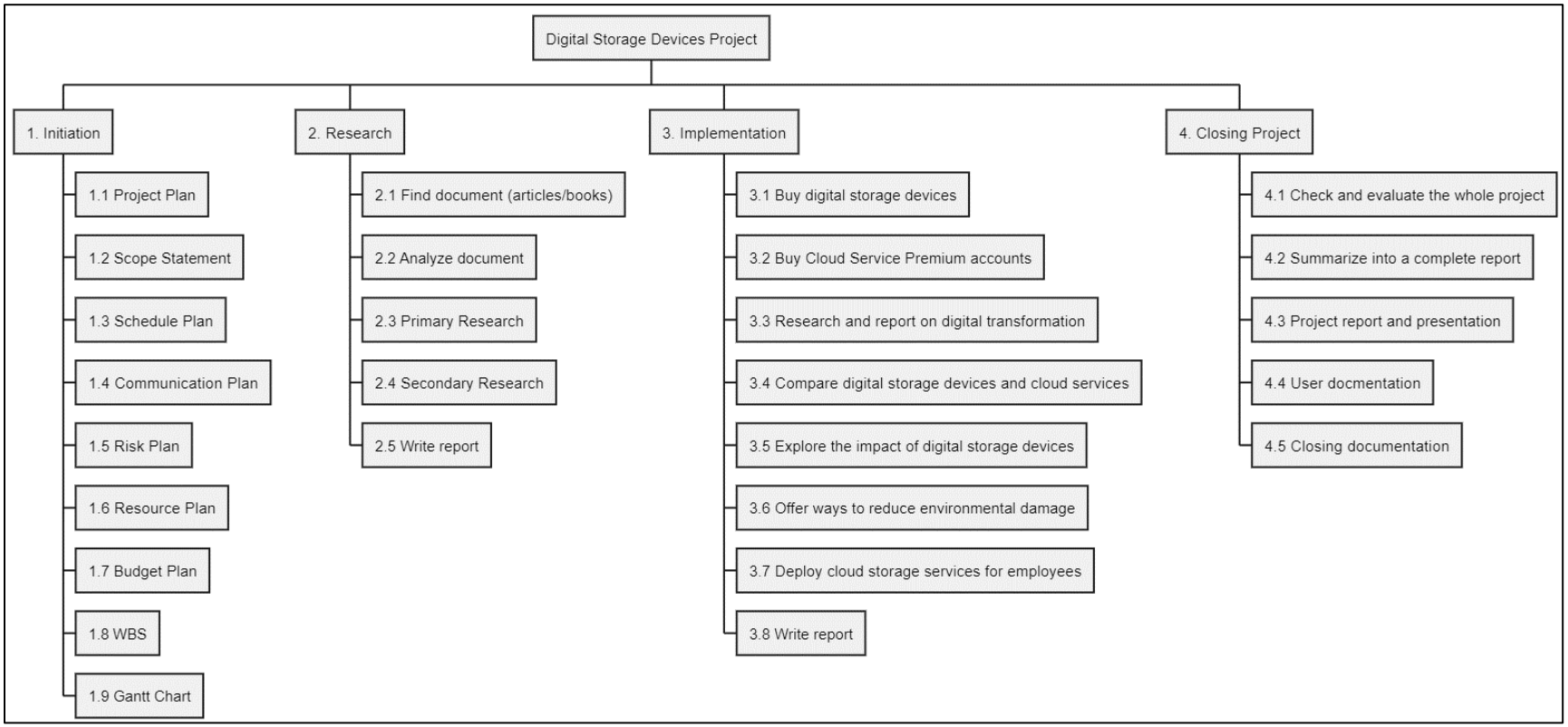


Figure 1: WBS

1.4.2 Gantt Chart

A Gantt chart is a horizontal bar chart developed as a production control tool in 1917 by Henry L. Gantt, an American engineer, and social scientist. One of the most common and effective methods of displaying activities (tasks or events) plotted against time is a Gantt chart, which is frequently used in project management. A list of the activities is located on the chart's left side, and a suitable time scale is located along the top. A bar is used to symbolize each activity, and the location and length of the bar correspond to the activity's beginning, middle, and finish dates (Gantt, 2022).

This allows you to see at a glance:

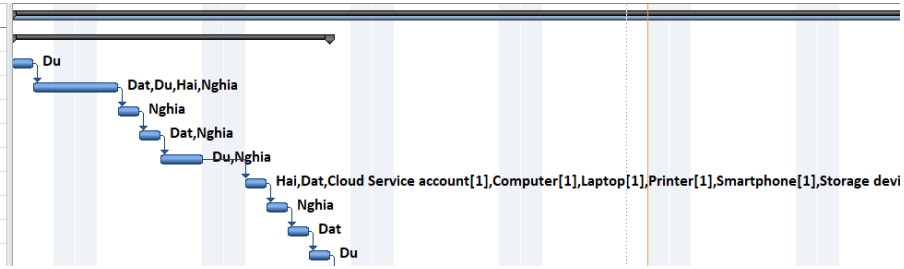
- What the various activities are
- When each activity begins and ends

- How long each activity is scheduled to last
- Where activities overlap with other activities, and by how much
- The start and end date of the whole project

The following is the Gantt Chart of our project:

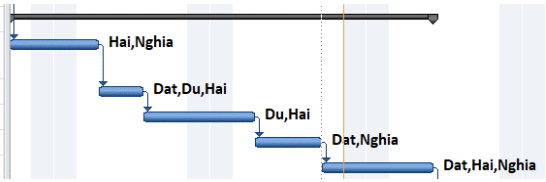
The initiation phase:

1	**Digital Storage Devices Project**	**56 days**	**Thu 9/15/22**	**Thu 12/1/22**	
2	**Initiation**	**11 days**	**Thu 9/15/22**	**Thu 9/29/22**	
3	Project Plan	1 day	Thu 9/15/22	Thu 9/15/22	
4	Scope Statement	2 days	Fri 9/16/22	Mon 9/19/22	3
5	Schedule plan	1 day	Tue 9/20/22	Tue 9/20/22	4
6	Communication plan	1 day	Wed 9/21/22	Wed 9/21/22	5
7	Risk plan	2 days	Thu 9/22/22	Fri 9/23/22	6
8	Resources plan	1 day	Mon 9/26/22	Mon 9/26/22	7
9	Budget plan	1 day	Tue 9/27/22	Tue 9/27/22	8
10	WBS	1 day	Wed 9/28/22	Wed 9/28/22	9
11	Gantt Chart	1 day	Thu 9/29/22	Thu 9/29/22	10



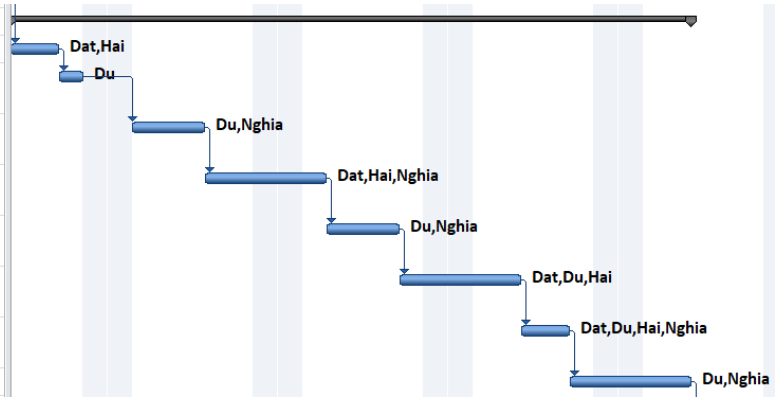
The research phase:

12	**Research**	**13 days**	**Fri 9/30/22**	**Tue 10/18/22**	
13	Find document (articles/books)	2 days	Fri 9/30/22	Mon 10/3/22	11
14	Analyze document	2 days	Tue 10/4/22	Wed 10/5/22	13
15	Primary research	3 days	Thu 10/6/22	Mon 10/10/22	14
16	Secondary research	3 days	Tue 10/11/22	Thu 10/13/22	15
17	Write report	3 days	Fri 10/14/22	Tue 10/18/22	16



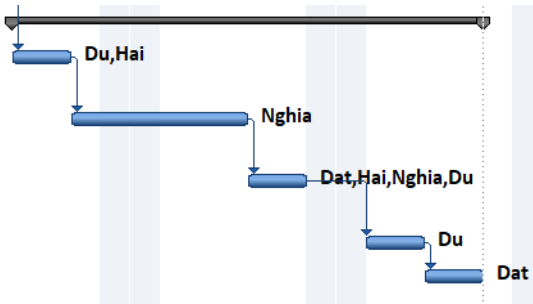
The implementation phase:

18	**Implementation**	**20 days**	**Wed 10/19/22**	**Tue 11/15/22**	
19	Buy digital storage devices	2 days	Wed 10/19/22	Thu 10/20/22	17
20	Buy Cloud Service Premium accounts	1 day	Fri 10/21/22	Fri 10/21/22	19
21	Research and report on digital transformation	3 days	Mon 10/24/22	Wed 10/26/22	20
22	Compare digital storage devices and cloud services	3 days	Thu 10/27/22	Mon 10/31/22	21
23	Explore the impact of digital storage devices	3 days	Tue 11/1/22	Thu 11/3/22	22
24	Offer ways to reduce environmental damage	3 days	Fri 11/4/22	Tue 11/8/22	23
25	Deploy cloud storage services for employees	2 days	Wed 11/9/22	Thu 11/10/22	24
26	Write report	3 days	Fri 11/11/22	Tue 11/15/22	25



The closing phase:

27	**Closing Project**	**12 days**	**Wed 11/16/22**	**Thu 12/1/22**	
28	Check and evaluate the whole project	2 days	Wed 11/16/22	Thu 11/17/22	26
29	Summarize into a complete report	4 days	Fri 11/18/22	Wed 11/23/22	28
30	Project report and presentation	2 days	Thu 11/24/22	Fri 11/25/22	29
31	User docmentation	2 days	Mon 11/28/22	Tue 11/29/22	30
32	Closing documentation	2 days	Wed 11/30/22	Thu 12/1/22	31



1.5 Produce a comprehensive project management plan, milestone schedule and project schedule for monitoring and completing the aims and objectives of the project

I utilized Microsoft Project, a project management tool, in this project to increase the effectiveness of project management. This program is used to create the Gantt charts seen in Figure 1.4.2, the detailed schedules, and the projected costs displayed in Tables 2 and 3. It is also simpler to track project progress with this program.

1.5.1 Microsoft Project

According to Megan Keup (2022), Microsoft Project is a project management tool used for scheduling, planning, managing resources, and timekeeping. For project management experts, it contains tools like Gantt charts, kanban boards, and project calendars. The current official name of the program is Microsoft Project, however, it is also referred to as MS Project or Project Professional. Project Online, Project Server, and Project for the Web are some further Microsoft Project products with related names.

Microsoft Project Features:

- **Project Planning:** Use Gantt charts and kanban boards to plan work. Microsoft Project's planning features are much more robust than those of Microsoft Planner.
- **Communication & Collaboration:** Teams can work together on projects. Alternatively, you can use Microsoft Teams for a more cost-effective communication solution.
- **Coauthoring:** Stakeholders and team members work together to edit and update task lists and schedules.
- **Reporting:** Pre-built reports that can track progress, resources, programs, and portfolios.
- **Roadmap:** Track programs and project portfolios.
- **Timesheets:** Collect project and non-project time for payroll and invoicing.

- **Resource Management:** Manage resources by requesting and assigning tasks.

Advantages of Microsoft Project:

- One of the biggest pros is that it's a Microsoft product so it integrates with the company's other products including Office 365, Skype and Sharepoint.
- It has a similar interface to other MS products.
- It's been around for a long time, and its features have developed over time.
- It's part of Microsoft and has the reliability and support that's associated with an established company.
- It has financial management tools that help project managers with estimating budgets.
- It can be licensed as a desktop application. While this might make it seem more like a dinosaur, there are still organizations that prefer this over a cloud solution.
- It has templates to help users get started, which saves time.

Disadvanatges of Microsoft Project:

- Despite being a cloud-based solution, MS Project Online has a number of drawbacks. Even with Sharepoint, which is built to utilize the cloud, MS Project has significant cloud-related constraints.
- It's challenging to use and learn. Before project managers and their teams feel comfortable utilizing the software, a significant amount of time, effort, and even intense training must be put into it. The project will take longer to implement as a result.
- It is pricey. Again, the costs shown above are per person, per month. As you purchase licenses for team members who need access to the program in order to utilize its collaborative capabilities, this quickly adds up. If not, it's more of a pricey tool reserved just for the project manager.
- It is difficult to share. As was already noted, files are saved in the proprietary MPP format, making them inaccessible to people who don't have MS Project. If the program were less expensive, this would be less of an issue, but it adds up if you need an MS Project license in order to open an MPP file. Sharing project files is made very difficult by these numerous needless obstacles.

1.5.2 Monitoring

Project monitoring entails keeping track of a project's metrics, development, and related duties to make sure everything is finished on schedule, within budget, and in accordance with the project's specifications and standards. Roadblocks or other difficulties that could occur throughout the course of the project's execution are also recognized and identified, and steps are taken to address these concerns (Dara Fontein, 2021).

The following table will show the progress monitoring of our project from 09/15/2022 to 12/01/2022:

Table 4: Monitoring progress

No.	Task Name	Duration	Start	Finish	Predecessors	% Complete
1	**Digital Storage Devices Project**	**56 days**	**Thu 9/15/22**	**Thu 12/1/22**		**45%**
2	**Initiation**	**11 days**	**Thu 9/15/22**	**Thu 9/29/22**		**100%**
3	Project Plan	1 day	Thu 9/15/22	Thu 9/15/22		100%
4	Scope Statement	2 days	Fri 9/16/22	Mon 9/19/22	3	100%
5	Schedule plan	1 day	Tue 9/20/22	Tue 9/20/22	4	100%
6	Communication plan	1 day	Wed 9/21/22	Wed 9/21/22	5	100%
7	Risk plan	2 days	Thu 9/22/22	Fri 9/23/22	6	100%
8	Resources plan	1 day	Mon 9/26/22	Mon 9/26/22	7	100%
9	Budget plan	1 day	Tue 9/27/22	Tue 9/27/22	8	100%
10	WBS	1 day	Wed 9/28/22	Wed 9/28/22	9	100%
11	Gantt Chart	1 day	Thu 9/29/22	Thu 9/29/22	10	100%
12	**Research**	**13 days**	**Fri 9/30/22**	**Tue 10/18/22**		**100%**
13	Find document (articles/books)	2 days	Fri 9/30/22	Mon 10/3/22	11	100%
14	Analyze document	2 days	Tue 10/4/22	Wed 10/5/22	13	100%
15	Primary research	3 days	Thu 10/6/22	Mon 10/10/22	14	100%
16	Secondary research	3 days	Tue 10/11/22	Thu 10/13/22	15	100%
17	Write report	3 days	Fri 10/14/22	Tue 10/18/22	16	100%
18	**Implementation**	**20 days**	**Wed 10/19/22**	**Tue 11/15/22**		**10%**
19	Buy digital storage devices	2 days	Wed 10/19/22	Thu 10/20/22	17	100%

20	Buy Cloud Service Premium accounts	1 day	Fri 10/21/22	Fri 10/21/22	19	0%
21	Research and report on digital transformation	3 days	Mon 10/24/22	Wed 10/26/22	20	0%
22	Compare digital storage devices and cloud services	3 days	Thu 10/27/22	Mon 10/31/22	21	0%
23	Explore the impact of digital storage devices	3 days	Tue 11/1/22	Thu 11/3/22	22	0%
24	Offer ways to reduce environmental damage	3 days	Fri 11/4/22	Tue 11/8/22	23	0%
25	Deploy cloud storage services for employees	2 days	Wed 11/9/22	Thu 11/10/22	24	0%
26	Write report	3 days	Fri 11/11/22	Tue 11/15/22	25	0%
27	**Closing Project**	**12 days**	**Wed 11/16/22**	**Thu 12/1/22**		**0%**
28	Check and evaluate the whole project	2 days	Wed 11/16/22	Thu 11/17/22	26	0%
29	Summarize into a complete report	4 days	Fri 11/18/22	Wed 11/23/22	28	0%
30	Project report and presentation	2 days	Thu 11/24/22	Fri 11/25/22	29	0%
31	User docmentation	2 days	Mon 11/28/22	Tue 11/29/22	30	0%
32	Closing documentation	2 days	Wed 11/30/22	Thu 12/1/22	31	0%

1.5.3 Critical Path Method

The critical path method (CPM) is a method for determining schedule flexibility and identifying activities required for project completion. The longest series of tasks that must be completed on schedule for the project to be concluded is known as a critical path in project management. The rest of the project will be delayed if important activities are delayed. Finding the most crucial activities in the project timetable, determining task dependencies, and computing task durations are the core tenets of CPM (Team Asana, 2021).

The following diagram is the CPM for our project:

Figure 2: CPM

2 Research

2.1 Primary Reasearch

2.1.1 Qualitative and quantitative research methods

Qualitative research methods:

In qualitative research, non-numerical data (such as text, video, or audio) are gathered and analyzed to better comprehend concepts, beliefs, or experiences. It may be applied to get a comprehensive understanding of a situation or to come up with fresh research concepts. Quantitative research involves gathering and analyzing numerical data for statistical analysis. Qualitative research does not. In fields like anthropology, sociology, education, the health sciences, history, etc., qualitative research is frequently employed in the humanities and social sciences (Pritha Bhandari, 2020).

The goal of qualitative research is to gain a thorough knowledge of social phenomena in their natural environment. It depends on the firsthand experiences of people as meaning-making agents in their daily lives and concentrates on the "why" rather than the "what" of social occurrences. For the study of human phenomena, qualitative researchers employ a variety of systems of inquiry, such as biography, case study, historical analysis, discourse analysis, ethnography, grounded theory, and phenomenology, as opposed to logical and statistical methods. Additionally, the social sciences are where qualitative methods first emerged. These strategies are "mainly concerned with enhancing comprehension of a substantive topic, rather than giving an explanation for it," according to Berndtsson et al. In the discipline of information science, qualitative methods which include techniques like case studies and surveys are more prevalent (Christian W. Dawson, 2009).

According to Pritha Bhandari (2020), each of the research approaches involves using one or more data collection methods. These are some of the most common qualitative methods:

- **Observations:** recording what you have seen, heard, or encountered in detailed field notes.
- **Interviews:** personally asking people questions in one-on-one conversations.
- **Focus groups:** asking questions and generating discussion among a group of people.
- **Surveys:** distributing questionnaires with open-ended questions.
- **Secondary research:** collecting existing data in the form of texts, images, audio or video recordings, etc.

Quantitative research methods:

In the social sciences, quantitative methodology dominates the research framework. It alludes to a collection of methods, approaches, and presumptions used to investigate numerical patterns in order to examine psychological, social, and economic phenomena. Numerous numerical data are gathered during quantitative research. While other times the numeric structure is enforced, certain numerical data is quantitative by

nature. Researchers can perform basic to more complex statistical studies that aggregate the data, illustrate linkages among the data, or compare across aggregated data thanks to the acquisition of quantitative data. In contrast to qualitative research, quantitative research use procedures including surveys, systematic observations, and experiments (Mary Brydon-Miller & David Coghlan, 2014).

The goal of quantitative analytic techniques is to quantify risks and their consequences. They frequently look at the potential influence on project costs and time. As an alternative, each detected risk may receive a rating number from the review process. The importance that risk should be given in terms of managerial focus and spending on preventative actions is indicated by the ranking numbers. Despite the fact that all quantitative approaches yield actual numbers, they may offer the impression of precision. It must be kept in mind that the outcomes are dependent on guesses, presumptions, and human judgment. It's possible that the evaluations made by the contributors are fundamentally erroneous, incorrect, or just too complex for anybody to make with any degree of accuracy (Dennis Lock, 2017).

These are some of the most common quantitative methods:

- **Surveys:** List of closed or multiple choice questions that is distributed to a sample (online, in person, or over the phone).
- **Experiments:** Situation in which variables are controlled and manipulated to establish cause-and-effect relationships.
- **Observations:** Observing subjects in a natural environment where variables can't be controlled.

The differences between quantitative and qualitative research:

Quantitative and qualitative research use different research methods to collect and analyze data, and they allow you to answer different kinds of research questions:

Table 5: Qualitative vs Quantitative

Qualitative	Quantitative
Emphasis on understanding	Emphasis on testing and verification
Focus on understanding from respondent's point of view	Focus on facts or reasons for social events
Focuses on exploring ideas and formulating a theory or hypothesis	Focuses on testing theories and hypotheses
Interpretation and rational approach	Logical and critical approach
Observations and measurements in natural settings	Controlled measurement
Subjective "insider view" and closeness to data	Objective "outsider view" distant from data
Explorative orientation	Hypothetical-deductive; focus on hypothesis/theory testing
Process oriented	Result oriented
Closed (multiple choice) questions	Open-ended questions
Holistic perspective	Particularistic and analytical

Generalization by comparison of properties and contexts of individual organism	Generalization by population membership

2.1.2 Overview about Interview/Survey

Interview questions:

1. What do you understand about digital transformation?
2. Have you ever tried using popular digital storage devices? If yes, what are they?
3. What do you think about the impact of digital storage devices on the environment?
4. What do you think we should do to reduce the damage of digital storage devices to the environment?
5. Do you think we should use cloud storage services instead of using digital storage devices? Give your reason.
6. What other digital devices do you think also affect the environment?

Summary interview question: With the interview questions above, we may collect user data on project-related issues to assist us in research and implementation. With the first question, we will know the existing knowledge of users about digital transformation. The second question will know what users have used digital storage devices and what they are. The third question tells us about users' thoughts on the impact of digital storage devices on the environment. The fourth question will ask users' thoughts on ways to reduce the harmful effects of digital storage devices on the environment. For question number 5, users will provide their opinions on using cloud storage as an alternative to digital storage and their reasons. The last question will indicate which other digital devices also affect the environment in the opinion of the user.

Survey questions:

Table 6: Survey questions

Survey questions	Answer				
	0	>1	>5	>10	>15
How many digital storage devices have you used?					
	Extremely unaffected	Unaffected	Neutral	Affected	Extremely affected
How do you think digital storage devices affect the environment?					
	Very dissatisfied	Dissatisfied	Neither satisfied nor dissatisfied	Satisfied	Very satisfied
How satisfied are you with today's digital storage devices?					

How satisfied are you with today's cloud storage services?					
	Strongly disagree	Disagree	Neither agree or disagree	Agree	Strongly agree
Do you think we need to replace digital storage devices with cloud storage services?					
Do you agree that we should intervene immediately to reduce the harmful effects of digital storage devices on the environment?					

Sumary survey questions: Similar to the interview questions in the previous section, these survey questions will also help us collect data for research and project development. The first question will give us the predicted number of digital storage devices that users have used. The second question will provide information about the influence of digital storage devices on users' opinions. With questions number three and four, we will know the level of user satisfaction with the use of digital storage devices and cloud storage services. The question number five will tell us what users think about using cloud storage instead of digital storage devices. The last question we will collect users' opinions on whether to intervene to reduce the damage of digital storage devices to the environment.

2.1.3 Evaluation about interview and survey

Through qualitative and quantitative research methods using data collection methods such as surveys and interview questions above, we are able to collect data from people to provide and support for our research. This collected data helps us to orient and adjust the development goals of the project. In addition, we may gain users' opinions and knowledge about the impact of digital storage devices on the environment, from which we can develop user-friendly documentations.

2.2 Secondary research

2.2.1 Overview

The secondary research uses information that has already been gathered by another party. This kind of information is referred to as "past data," and it is often available through previous studies, public documents, and different online and offline venues. In conclusion, secondary research entails reexamining, analyzing, or revising earlier findings. It is always the responsibility of the researcher to explain how this historical material has influenced their current work. Secondary research is simpler than primary research, in part because the researcher is less directly involved in the data collection process. Additionally, secondary research takes up less time and costs less money because you don't have to pay for any additional study-related expenses or compensate participants for their time. Textbooks, encyclopedias, news stories, review papers, and meta analyses are typical forms of secondary research. Authors may use data from scholarly

publications, official records, statistical databases, and historical documents while performing secondary research (OxbridgeEssays, 2021).

Advantages of the secondary research:

- **Inexpensive:** Conducting secondary research is much cheaper than doing primary research
- **Saves time:** Secondary research takes much less time than primary research
- **Accessibility:** Secondary data is usually easily accessible from online sources.
- **Large scope of data:** You can rely on immensely large data sets that somebody else has collected
- **Professionally collected data:** Secondary data has been collected by researchers with years of experience

Disadvantages of the secondary research:

- **Inappropriateness:** Secondary data may not be fully appropriate for your research purposes
- **Wrong format:** Secondary data may have a different format than you require
- **May not answer your research question:** Secondary data was collected with a different research question in mind
- **Lack of control over the quality of data:** Secondary data may lack reliability and validity, which is beyond your control
- **Lack of sufficient information:** Original authors may not have provided sufficient information on various research aspects

2.2.2 List of articles

Table 7: List of articles

No.	Title	Authors	Year	Type
1	Digital Transformation and Environmental Sustainability: A Review and Research Agenda	Hangjung Zo, Ananth Chiravuri, Abdul Karim Feroz	2021	Article
2	The Impact of Digital Transformation on Environmental Sustainability	Thanh Cong Truong	2022	Article
3	Indirect Effects of the Digital Transformation on Environmental Sustainability: Methodological Challenges in Assessing the Greenhouse Gas Abatement Potential of ICT	Jan C. T. Bieser, Lorenz M. Hilty	2018	Article

2.2.3 Summarize about articles

Digital Transformation and Environmental Sustainability: A Review and Research Agenda

Digital technology is the driving force behind the process of "digital transformation," which causes upheavals inside enterprises. Global markets and industries have been disrupted as a result of digital transformation, which has sparked new business models. The main drivers of digital transformation are IoT, big data analytics, cloud computing, mobile technology, and artificial intelligence. Environmental and health gains are made via digital technology. The sustainability triangle, or environmental sustainability, is anticipated to suffer from digital transformation. The implications of digital changes on environmental sustainability are identified and categorized in this article into four categories: waste management, pollution control, sustainable production, and urban sustainability. The greatest challenge continues to be safeguarding the environment from pollution and resource deterioration. Measurement and management of challenges connected to environmental sustainability are changing as a result of digital technology. Protecting the environment from pollution and resource deterioration is one of the goals of this research. Using the framework, the author discovered that significant disruptions brought on by the digital era affect sustainable manufacturing, waste management, pollution control, and urban sustainability.

The Impact of Digital Transformation on Environmental Sustainability

The usage of digital technologies like artificial intelligence (AI), big data analytics, the Internet of things (IoT), blockchain, and other technologies is causing dramatic changes in every element of society, businesses, and industries. There is a complicated link between digital transformation and environmental sustainability, and there is disagreement among scientists as to whether digital transformation promotes or impedes environmental sustainability. The evaluations of the effects of digital products on the environment were reviewed by the authors. The term "digital transformation" refers to the substantial changes brought about by the use of digital technologies like artificial intelligence (AI), big data analytics, the Internet of things (IoT), blockchain, and other technologies in every sphere of society, organizations, and industries. Researchers assert that digital transformation might aid in reducing the effects of climate change in urban areas. The purpose of this study is to determine how digital transformations affect the sustainability of the environment. Other research has concentrated on the indirect effects of ICTs on the environment. The publication omitted clear mention of the negative implications of digitization. The current study set out to investigate how digital change affects environmental sustainability. This study looked at how digital technology may reduce environmental pollution and improve the effectiveness of waste collection and management. This study supports the notion that environmental sustainability goals may be greatly advanced through digital transformation. However, it also presents a number of difficulties that demand a deeper analysis. To sum up, the authors think that their research sets the foundation for further investigation in the area of environmental sustainability.

Indirect Effects of the Digital Transformation on Environmental Sustainability: Methodological Challenges in Assessing the Greenhouse Gas Abatement Potential of ICT

The digital transition affects greenhouse gas (GHG) emissions both directly and indirectly. The creation, usage, and disposal of information and communication technology (ICT) hardware have direct implications. Indirect effects include changes to patterns of production and consumption in other domains. Studies that measure both impacts frequently come to the conclusion that net effects can result in a sizable decrease in GHG emissions. Information processing, transmission, and storage technologies have converged as a result of digital electronics. Digitalization has two effects on GHG emissions: On the one hand, a rising quantity of ICT gear is created, utilized, and eventually disposed of away while being powered by energy. ICT, on the other hand, has an impact on production and consumption habits, which might have a variety of negative

repercussions (indirect effects). Numerous studies have been done recently to estimate both direct and indirect impacts. This research typically come to the conclusion that indirect effects outweigh direct impacts. The authors come to the conclusion that special regulations are required to maximize ICT's good potential while minimizing its negative ones.The Swiss economy might save up to 11.32 Mt CO2e annually because to ICT (optimistic scenario). This result must be evaluated carefully, though, as there are significant methodological issues with studies of the indirect environmental consequences of ICT. ICT won't help Switzerland maintain its climate if no steps are made and anticipated effects can't be fulfilled. A further danger is posed by rebound effects, which raise demands in response to cost reductions. The outcome of a study is significantly influenced by how it approaches these difficulties. The ICT industry has a unique chance to help other industries implement ICT-based ("smart") low-carbon solutions in terms of both technology and business models, as well as a social duty as a result of this opportunity.

2.2.4 Evaluation about your understanding

After studying the above articles, I have a better understanding of digital transformation specifically digital transformation refers to the unprecedented disruptions in society, industry, and organizations stimulated by advances in digital technologies such as artificial intelligence, big data analytics, cloud computing, and the Internet of Things (IoT). Furthermore, I have clearly understood the impacts of digital transformation on environmental sustainability, including both positive and negative effects. Additionally, the articles highlight the transformations that preserve the environment in three main areas: waste management and handling, pollution prevention, and control, and sustainable resource management. These papers identify the disruptions driven by digital transformation in the environmental sustainability domain through a systematic literature review. The results present a framework that outlines the transformations in four key areas: pollution control, waste management, sustainable production, and urban sustainability. Moreover, I also know more about digital transformation has direct and indirect effects on greenhouse gas (GHG) emissions. Direct effects are caused by the production, use, and disposal of information and communication technology (ICT) hardware. Indirect effects include changes to patterns of production and consumption in other domains.

However, besides the knowledge that I have gained, there are a few points that I still do not grasp and understand. Specifically, I cannot clearly understand the research methods that the authors have applied including statistical tables and calculation methods. Also, because the authors use too many academic words and complex sentences, I still do not fully understand the idea they want to express. Finally, I still do not fully understand the solutions proposed by the authors to reduce the environmental harm of digital devices in particular and the digital transformation process in general.

If I had more time, I would continue to research and learn more about the articles above to find and understand the ideas the authors wanted to interpret. In addition, I will also learn more about other books or articles about the impact of digital transformation on the environment. From there, I can confirm the point that I have made for the project as well as in this report. Finally, if I had more time, I would apply and present statistical methods of data and scientific computation as evidence to demonstrate to the reader that the effects of digital storage devices on the environment that I have given in my project are correct.

3 References

Adeaca Corp, 2021. *Adeaca.* [Online]
Available at: https://www.adeaca.com/blog/faq-items/what-is-project-execution/
[Accessed 12 October 2022].

Adrienne Watt, 2014. *Project Management – 2nd Edition.* 2nd ed. s.l.:BCcampus.

Cassie Bottorff & Alana Rudder, 2022. *Forbes.* [Online]
Available at: https://www.forbes.com/advisor/business/project-management-plan/#:~:text=A%20project%20management%20plan%20is%20a%20set%20of%20documents%20that,involved%20in%20a%20project%27s%20completion.
[Accessed 12 October 2022].

Christian W. Dawson, 2009. *Project in Computing and Information Systems.* 2nd ed. s.l.:Pearson Education.

Dara Fontein, 2021. *Unito.* [Online]
Available at: https://unito.io/blog/project-monitoring/#:~:text=But%2C%20what%20exactly%20is%20project,to%20project%20requirements%20and%20standards.
[Accessed 15 October 2022].

Dennis Lock, 2017. *Project Management.* 9th ed. s.l.:Gower Publishing Limited.

Gantt, 2022. *Gantt.* [Online]
Available at: https://www.gantt.com/
[Accessed 15 October 2022].

Imagine Projects, 2021. *Imagine Projects.* [Online]
Available at: https://imagineprojects.co.uk/project-inception-phase/
[Accessed 12 October 2022].

Indeed Editorial Team, 2020. *Indeed.* [Online]
Available at: https://www.indeed.com/career-advice/career-development/what-is-scope-in-project-management
[Accessed 12 October 2022].

Mary Brydon-Miller & David Coghlan, 2014. *The SAGE Encyclopedia of Action Research.* 1st ed. London: SAGE Publications Ltd.

Mary K. Pratt, 2018. *TechTarget.* [Online]
Available at: https://www.techtarget.com/searchcio/definition/project-management
[Accessed 12 October 2022].

Megan Keup, 2022. *ProjectManager.* [Online]
Available at: https://www.projectmanager.com/blog/what-is-microsoft-project
[Accessed 15 October 2022].

OxbridgeEssays, 2021. *OxbridgeEssays.* [Online]
Available at: https://www.oxbridgeessays.com/blog/how-to-dissertation-secondary-research-4-steps/#thebasics
[Accessed 17 October 2022].

Pritha Bhandari, 2020. *Scribbr.* [Online]
Available at: https://www.scribbr.com/methodology/qualitative-research/#:~:text=Revised%20on%20October%2010%2C%202022,generate%20new%20ideas%20for%20research.
[Accessed 16 October 2022].

Shawn Dickerson, 2021. *ProjectManager.* [Online]
Available at: https://www.projectmanager.com/blog/project-time-management-strategies#:~:text=Project%20time%20management%20is%20the,produce%20the%20project%27s%20final%20deliverable.
[Accessed 12 October 2022].

Simplilearn, 2022. *Simplilearn.* [Online]
Available at: https://www.simplilearn.com/how-to-close-a-project-article
[Accessed 12 October 2022].

Team Asana, 2021. *Asana.* [Online]
Available at: https://asana.com/resources/critical-path-method
[Accessed 16 October 2022].

Techopedia, 2012. *Techopedia.* [Online]
Available at: https://www.techopedia.com/definition/14005/project-planning
[Accessed 12 October 2022].

Tim Stobierski, 2019. *Northeastern University.* [Online]
Available at: https://www.northeastern.edu/graduate/blog/cost-estimation-in-project-management/
[Accessed 13 October 2022].

Tuorialspoint, 2022. *TuorialSpoint.* [Online]
Available at:
https://www.tutorialspoint.com/software_engineering/software_project_management.htm#:~:text=A%20Software%20Project%20is%20the,to%20achieve%20intended%20software%20product.
[Accessed 12 October 2022].